



# 2026

## Chemistry

Paper 4 — Organic Chemistry · Types of Reactions · Balancing Equations

**Time allowed:** 45 minutes

**Total marks:** 50

**Index Number:** \_\_\_\_\_

### Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer **all** questions in **all three sections**.
3. Section A: Choose **one** answer for each question. Tick the box next to your chosen answer.
4. Section B & C: Write your answers in the spaces provided.
5. Use a blue or black pen. Diagrams may be drawn in pencil.
6. All working must be shown for calculations.
7. The number of marks is given in brackets [ ] at the end of each question.

| Section      | Marks Available | Marks Awarded |
|--------------|-----------------|---------------|
| A            | 15              |               |
| B            | 20              |               |
| C            | 15              |               |
| <b>Total</b> | <b>50</b>       |               |

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**SECTION A — Multiple Choice [15 marks] Answer all 15 questions. Tick ONE box per question.**

- 1 The general formula of the **alkane** homologous series is [1]
- ☐ A  $C_n H_{2n}$
- ☐ B  $C_n H_{2n+2}$
- ☐ C  $C_n H_{2n-2}$
- ☐ D  $C_n H_n$
- 2 How many carbon atoms are in one molecule of methane? [1]
- ☐ A 1
- ☐ B 2
- ☐ C 3
- ☐ D 4
- 3 Ethene ( $C_2H_4$ ) belongs to the homologous series called [1]
- ☐ A alkanes
- ☐ B alkenes
- ☐ C alcohols
- ☐ D carboxylic acids
- 4 The functional group present in all alcohols is [1]
- ☐ A  $-COOH$
- ☐ B  $-CHO$
- ☐ C  $-OH$
- ☐ D  $-C=C-$
- 5 Alkanes react with chlorine in the presence of ultraviolet light by [1]
- ☐ A addition reaction
- ☐ B substitution reaction
- ☐ C neutralisation
- ☐ D polymerisation
- 6 Alkenes decolourise bromine water because they undergo [1]
- ☐ A substitution reactions
- ☐ B combustion reactions
- ☐ C addition reactions across the  $C=C$  double bond

☐ D cracking reactions

7 The complete combustion of a hydrocarbon always produces

[1]

- ☐ A carbon monoxide and water
- ☐ B carbon dioxide and water
- ☐ C carbon and steam
- ☐ D carbon dioxide and hydrogen

8 The reaction between an alcohol and a carboxylic acid (in the presence of an acid catalyst) produces

[1]

- ☐ A a polymer and water
- ☐ B an alkene and water
- ☐ C an aldehyde and hydrogen
- ☐ D an ester and water

9 Cracking is an industrial process used to

[1]

- ☐ A break down large, less useful hydrocarbon molecules into smaller, more useful ones
- ☐ B combine small hydrocarbon molecules into larger ones
- ☐ C remove sulfur impurities from crude oil
- ☐ D separate fractions from crude oil by boiling point

10 The IUPAC name of the alkane with the molecular formula  $C_3H_8$  is

[1]

- ☐ A methane
- ☐ B propane
- ☐ C butane
- ☐ D ethane

11 The reaction  $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$  is an example of

[1]

- ☐ A addition
- ☐ B substitution
- ☐ C combustion
- ☐ D polymerisation

12 Ethanol ( $C_2H_5OH$ ) belongs to which homologous series?

[1]

- ☐ A alkanes
- ☐ B alkenes
- ☐ C alcohols
- ☐ D carboxylic acids

- 13 In a correctly balanced chemical equation, which of the following must be equal on both sides? [1]
- ☐ A The number of atoms of each element (mass must be conserved)
  - ☐ B The number of molecules
  - ☐ C The number of moles of gas produced
  - ☐ D The total number of compounds
- 14 The complete combustion of propane ( $\text{C}_3\text{H}_8$ ) is represented by which balanced equation? [1]
- ☐ A  $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
  - ☐ B  $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
  - ☐ C  $\text{C}_3\text{H}_8 + 3\text{O}_2 \rightarrow 3\text{CO}_2 + \text{H}_2\text{O}$
  - ☐ D  $2\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 6\text{CO} + 8\text{H}_2\text{O}$
- 15 Addition polymers are made from [1]
- ☐ A alkane monomers by substitution
  - ☐ B alkene monomers joining together, with no other product
  - ☐ C alcohol monomers reacting with an acid catalyst
  - ☐ D two different monomers with water as a by-product

**SECTION B — Short Structured Questions [20 marks] Answer all questions.**

**B1 The Alkane Homologous Series**

The table below shows some members of the alkane homologous series.

| Name    | Molecular formula              | Structural formula (displayed)                                  | Boiling point / °C |
|---------|--------------------------------|-----------------------------------------------------------------|--------------------|
| Methane | CH <sub>4</sub>                | CH <sub>4</sub>                                                 | -162               |
| Ethane  | C <sub>2</sub> H <sub>6</sub>  | CH <sub>3</sub> CH <sub>3</sub>                                 | -89                |
| Propane | C <sub>3</sub> H <sub>8</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>3</sub>                 | -42                |
| Butane  | C <sub>4</sub> H <sub>10</sub> | CH <sub>3</sub> (CH <sub>2</sub> ) <sub>2</sub> CH <sub>3</sub> | -1                 |
| Pentane | C <sub>5</sub> H <sub>12</sub> | CH <sub>3</sub> (CH <sub>2</sub> ) <sub>3</sub> CH <sub>3</sub> | 36                 |

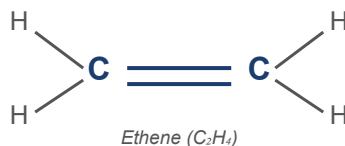
- (a) State the molecular formula of the **sixth** member of the alkane series (hexane). [1]

- (b) Describe the trend in boiling point as the chain length increases. Suggest a reason for this trend. [2]

- (c) State **two** features that all members of a homologous series share (apart from general formula). [2]

**B2 Reactions of Alkenes**

The structural formula of ethene is shown below.



- (a) Ethene reacts with bromine ( $Br_2$ ). Name the type of reaction and write the structural equation for the product formed. [2]

**Type of reaction:**

**Product (name and formula):**

- (b) State the observation when ethene gas is bubbled through bromine water. [1]

- (c) Ethene can undergo addition polymerisation to form poly(ethene). Draw a section of the polymer chain showing **three** repeat units. [2]

### B3 Balancing Chemical Equations

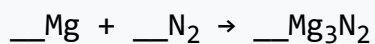
Balance each of the following equations by inserting the correct coefficients. Do not change the formulae of the compounds.

- (a)

The combustion of ethane:  $\text{---}C_2H_6 + \text{---}O_2 \rightarrow \text{---}CO_2 + \text{---}H_2O$

- (b) The reaction of magnesium with nitrogen gas:

[1]



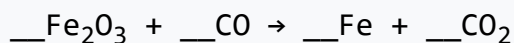
- (c) The thermal decomposition of calcium carbonate:

[1]



- (d) The reaction of iron(III) oxide with carbon monoxide during iron extraction:

[1]



#### B4 Combustion Reactions and Types of Reactions

- (a) Write the balanced equation for the **complete combustion** of methane ( $\text{CH}_4$ ).

[1]

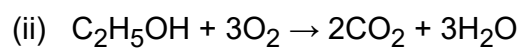
- (b) Write the balanced equation for the **incomplete combustion** of methane, where carbon monoxide ( $\text{CO}$ ) is produced instead of  $\text{CO}_2$ .

[1]

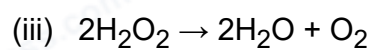
- (c) Classify each of the following reactions as one of: **combustion / substitution / addition / decomposition / neutralisation**.



[1]



[1]



[1]

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**SECTION C — Long Structured Questions [15 marks] Answer all questions.**

**C1 Organic Chemistry — Crude Oil, Cracking, and Functional Groups**

Crude oil is a mixture of hydrocarbons that can be separated by fractional distillation. The heavier fractions contain large hydrocarbon molecules that are less in demand. These undergo **cracking** to produce smaller, more useful molecules.

(a) State what is meant by the term *hydrocarbon*.

(b) Decane ( $C_{10}H_{22}$ ) can be cracked to give butane ( $C_4H_{10}$ ) and a mixture of alkenes. Write a balanced equation for one possible cracking of decane to give butane and two molecules of propene ( $C_3H_6$ ).

(c) State the conditions needed for thermal cracking.

(d) The table below shows four organic compounds. Complete the table by filling in the missing entries.

| Name          | Formula        | Homologous series | Functional group |
|---------------|----------------|-------------------|------------------|
| Ethanol       | $C_2H_5OH$     |                   |                  |
| Ethanoic acid | $CH_3COOH$     |                   |                  |
|               | $CH_3CH_2CH_3$ | Alkane            |                  |
| Propene       |                | Alkene            | $-C=C-$          |

**C2 Balancing Equations and Types of Reactions**

This question requires you to write and balance equations for various chemical reactions.

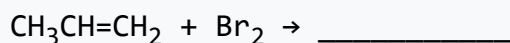
- (a) Write the **balanced symbol equation**, with state symbols, for the reaction of zinc with dilute sulfuric acid to produce zinc sulfate and hydrogen gas.

[2]

- (b) Write the balanced equation for the complete combustion of ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ). Include state symbols.

[2]

- (c) The following equation shows the addition of bromine to propene. Fill in the missing formula for the organic product.



Name the product and state

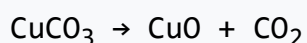
the type of reaction.

[2]

- (d) The thermal decomposition of copper(II) carbonate is represented by the unbalanced equation below. Balance it and state what type of

[1]

reaction this is.



# Mark Scheme — Paper 4

**For teacher/self-assessment use only.** For balanced equations: award 1 mark for correct formulae, 1 mark for correct balancing (unless stated otherwise). State symbols required only where the question asks for them. Structural formulae: accept any correct representation.

## Section A — Multiple Choice

| Q | Answer | Q  | Answer | Q  | Answer |
|---|--------|----|--------|----|--------|
| 1 | B      | 6  | C      | 11 | B      |
| 2 | A      | 7  | B      | 12 | C      |
| 3 | B      | 8  | D      | 13 | A      |
| 4 | C      | 9  | A      | 14 | B      |
| 5 | B      | 10 | B      | 15 | B      |

## Section B

| Question | Mark-worthy points                                                                                                                                                                       | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| B1(a)    | $C_6H_{14}$                                                                                                                                                                              | 1     |
| B1(b)    | Boiling point <u>increases</u> with chain length [1]; longer chains have stronger van der Waals (intermolecular) forces between molecules, so more energy is needed to separate them [1] | 2     |
| B1(c)    | Any 2: same general formula; differ by $CH_2$ successive members; similar chemical properties; gradual change in physical properties                                                     | 2     |
| B2(a)    | Addition [1]; product: 1,2-dibromoethane / $CH_2BrCH_2Br$ [1]                                                                                                                            | 2     |
| B2(b)    | Bromine water is decolourised (turns from brown/orange to colourless)                                                                                                                    | 1     |
| B2(c)    | Correct repeat unit $-(CH_2-CH_2)-$ [1]; three repeat units shown with brackets and bonds extending to chain [1]; (accept $-CH_2CH_2-$ notation)                                         | 2     |
| B3(a)    | $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$ [1 for correct formulae, 1 for 2,7,4,6]                                                                                                       | 2     |
| B3(b)    | $3Mg + N_2 \rightarrow Mg_3N_2$ (coefficients 3, 1, 1)                                                                                                                                   | 1     |
| B3(c)    | $CaCO_3 \rightarrow CaO + CO_2$ (already balanced — coefficients all 1; accept as-is)                                                                                                    | 1     |
| B3(d)    | $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (coefficients 1, 3, 2, 3)                                                                                                                        | 1     |
| B4(a)    | $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$                                                                                                                                                   | 1     |

| Question   | Mark-worthy points                                                        | Marks |
|------------|---------------------------------------------------------------------------|-------|
| B4(b)      | $2\text{CH}_4 + 3\text{O}_2 \rightarrow 2\text{CO} + 4\text{H}_2\text{O}$ | 1     |
| B4(c)(i)   | Addition                                                                  | 1     |
| B4(c)(ii)  | Combustion                                                                | 1     |
| B4(c)(iii) | Decomposition (thermal decomposition)                                     | 1     |

### Section C

| Question                         | Mark-worthy points                                                                                                                                                                                                                                                                              | Marks |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| C1(a)                            | A compound containing <u>carbon and hydrogen only</u> .                                                                                                                                                                                                                                         | 1     |
| C1(b)                            | $\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_4\text{H}_{10} + 2\text{C}_3\text{H}_6$ [1 for correct formulae, 1 for balanced: C: 4+6=10 ✓; H: 10+12=22 ✓]                                                                                                                                   | 2     |
| C1(c)                            | High temperature (400–900 °C) [1]; and either high pressure (thermal cracking) or a catalyst / silica or alumina catalyst (catalytic cracking) [1] — award 1 mark for temperature alone                                                                                                         | 1     |
| C1(d) — 1 mark each correct row: | Row 1: Alcohols; –OH [1]<br>Row 2: Carboxylic acids; –COOH [1]<br>Row 3: Propane; no functional group / C–C and C–H only [1]<br>Row 4: $\text{C}_3\text{H}_6$ [1]                                                                                                                               | 4     |
| C2(a)                            | $\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$ [1 for formulae, 1 for state symbols]                                                                                                                                             | 2     |
| C2(b)                            | $\text{C}_2\text{H}_5\text{OH(l)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)} + 3\text{H}_2\text{O(l)}$ [1 for balanced equation, 1 for state symbols]                                                                                                                           | 2     |
| C2(c)                            | Product: $\text{CH}_3\text{CHBrCH}_2\text{Br}$ (1,2-dibromopropane) [1];<br>Addition reaction [1]                                                                                                                                                                                               | 2     |
| C2(d)                            | Already balanced (coefficients all 1): $\text{CaCO}_3 \rightarrow \text{CuO} + \text{CO}_2$ — note: this is $\text{CuCO}_3 \rightarrow \text{CuO} + \text{CO}_2$ balanced as-is [1 for confirming balanced]; type: thermal decomposition [award mark even if student states decomposition only] | 1     |

### Key Balanced Equations — Summary (all four papers)

| Reaction                              | Balanced equation                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------|
| $\text{HCl} + \text{NaOH}$            | $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$                    |
| $\text{CaCO}_3 + \text{HCl}$          | $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ |
| $\text{CuO} + \text{H}_2\text{SO}_4$  | $\text{CuO} + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{H}_2\text{O}$        |
| $\text{ZnO} + \text{H}_2\text{SO}_4$  | $\text{ZnO} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2\text{O}$        |
| $2\text{H}_2\text{O}_2$ decomposition | $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$                       |
| $\text{CH}_4$ complete combustion     | $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$                  |

| Reaction                                             | Balanced equation                                                                            |
|------------------------------------------------------|----------------------------------------------------------------------------------------------|
| C <sub>2</sub> H <sub>6</sub> complete combustion    | $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$         |
| C <sub>3</sub> H <sub>8</sub> complete combustion    | $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$          |
| C <sub>2</sub> H <sub>5</sub> OH complete combustion | $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$ |
| Cracking decane                                      | $\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_4\text{H}_{10} + 2\text{C}_3\text{H}_6$     |
| Zn + H <sub>2</sub> SO <sub>4</sub>                  | $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$                   |
| Fe <sub>2</sub> O <sub>3</sub> + CO                  | $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$                   |
| Cathode (CuSO <sub>4</sub> , inert)                  | $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$                                         |
| Anode (CuSO <sub>4</sub> , inert)                    | $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$                     |
| Cathode (PbBr <sub>2</sub> )                         | $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$                                         |
| Ionic neutralisation                                 | $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$    |